

Read each question carefully, analyze the data provided on Canvas, and answer the questions based on your analysis. You are welcome to use class materials, and to search the Internet for definitions, but please do not ask specific questions from the exam sheet of anyone else, including AI – your answers should be the product of your own thinking, not the thinking of another person or a computer.

1. Cholesterol is a waxy, lipid-like substance that is synthesized by the liver. Some cholesterol is needed for normal body functioning, but an excessive amount of cholesterol in the blood can cause blockages to form in blood vessels, which in turn can cause heart attacks and strokes. Cholesterol comes in two forms, high-density lipoprotein (HDL), and low-density lipoprotein (LDL). LDL is considered “bad” cholesterol, because it is the one that forms blockages in blood vessels.

In addition to the cholesterol synthesized by the liver, we also ingest cholesterol in our diets when we eat animal-based foods. It seems likely that reducing cholesterol in our diets would lower blood levels. Evelyn decides to test whether patients with elevated cholesterol can lower their LDL levels just by eating the TLC diet, which is a low-cholesterol diet recommended by the National Institutes of Health. She selects 32 patients that all have elevated LDL levels, and randomly assigns half to the TLC diet. The remaining 16 patients continue their normal dietary practices as the controls. After one month, she measures the LDL levels in all of the patients. The data are in question1.xlsx, expressed as LDL concentrations, in mg/dL.

A. What is the null hypothesis that you will test to address whether diet lowers LDL level? Write this in symbolic notation (that is H_0 : ..., using the appropriate symbol for the mean, either μ or $\bar{x}$, and explain why you used the symbol you selected).

$H_0: \mu_{\text{Normal diet}} = \mu_{\text{TLC}}$

B. What statistical test will you use to test this null hypothesis? Give just the name of the procedure that tests for differences between the groups, not the tests that check assumptions.

A two-sample t-test (independent samples)

C. Test the assumptions of the procedure you selected for question B (the assumption tests are available in the analysis tool, use that rather than the tests in Descriptives).

Report test statistics and p-values for each, and write “yes” if you pass the assumption test, and “no” if you fail the assumption test.

	HOV (F statistic)	Normality (W statistic)
Test statistic	0.304	0.979
p-value	0.585	0.766
Pass?	Yes	Yes

D. Now test the null hypothesis, and report the results below:

t-value 3.87 d.f. 30 p <0.001 Reject or retain null? Reject

Normal diet group mean: 117 TLC group mean: 93.8

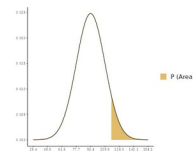
E. Interpret the results – did diet reduce LDL levels? How did you reach this conclusion from the p-value and group means?

Yes, the difference was significant, and the mean for TLC was lower than for normal diet, so eating the TLC diet reduced LDL levels.

F. Use the normal distribution to find each, and sketch the curve with the shaded region that gave you each probability:

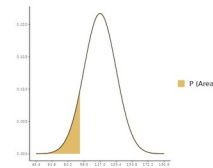
The probability that an individual patient in the TLC diet group would have an LDL level above the mean for the normal diet:

$p = 0.075$



The probability that a patient in the normal diet group would have an LDL level below the mean of the TLC diet:

$p = 0.104$ or 0.099 (depending on if you rounded the mean)



G. Does a significant reduction in LDL on the TLC diet mean that everyone on the TLC diet will have a low LDL level? How do your results from D and F address this question (which shows a difference on average, which addresses individual LDL values)?

No, the mean for the TLC group is lower than for the normal diet group, but some of the people on the normal diet have lower LDL than the mean of the TLC group, and some of the people on the TLC diet have higher LDL than the mean of the normal diet group.

2. Doctors recommend LDL levels at or below 100 mg/dL. To test if the diets are producing mean LDL levels higher than this, equivalent to this, or lower than this Evelyn reorganized the data with the normal diet in one column and TLC diet in another. The data are in file question2.xlsx.

A. To test if the means for each diet are equal to 100 mg/dL what kind of t-test will we need to use?

A one-sample t-test is used to compare means to numbers.

B. What is the null hypothesis for this test? Write it in symbolic notation (as before).

$H_0: \mu_{\text{Normal diet}} = 100$

$H_0: \mu_{\text{TLC}} = 100$

C. What assumptions are made by the t-test you are about to use? Have we already tested the assumptions needed for these data? If not, conduct any additional assumption tests and report the results here.

It assumes normal distributions of the data, which we tested in question 1 above, so we don't need to test it again.

D. Conduct the two separate tests, one for each column of data, and report the results:

Normal diet: t-value 3.79, d.f. 15, p 0.002

Do you reject the null or retain the null for the normal diet? Reject the null

TLC diet: t-value -1.53, d.f. 15, p 0.146

Do you reject the null or retain the null for the TLC diet? Retain the null

E. Interpret the results. Does the TLC diet produce LDL levels that are at 100 mg/dL, higher than 100 mg/dL, or below 100 mg/dL? What about the normal diet?

The normal diet has LDL levels above 100 mg/dL, and the TLC diet reduces LDL to levels equal to 100 mg/dL

F. Calculate the probability that a person on the TLC diet will have an LDL level above the recommended level of 100 mg/dL (use the normal distribution for this). What percentage of people on the TLC diet would be expected to have elevated LDL given this probability?

The probability is 0.350, which is 35% of the population that would be expected to have elevated LDL levels on the TLC diet.

3. It is possible that the TLC diet consistently reduces LDL level, but some individuals start with such high levels that they do not have their levels drop below 100 mg/dL. If this is the case, the effectiveness of the diet may be more easily seen by measuring LDL level before and after a group of patients uses the diet, instead of comparing a treatment and control group.

To check this, Evelyn does a new experiment in which a new group of 16 high LDL patients are selected, and all of them are put on the TLC diet for one month. Their LDL levels are measured at the beginning of the study, and then again at the end of the study. The data for this version of the study is in question3.xlsx.

A. In what way are the data in this experiment paired? What kind of t-test should be used given this?

The same people are measured before and after starting the TLC diet, so each person's before measurement is paired with their after measurement.

B. What is the null hypothesis for this paired analysis?

$H_0: \mu_{\text{Normal diet}} = \mu_{\text{TLC}}$ or

$H_0: \mu_{\text{Normal diet}} - \mu_{\text{TLC}} = 0$

C. Test the normality assumption for the differences between the before and after TLC values (this is available in the analysis too, use that rather than the test in Descriptives).

	Normality (W statistic)
Test statistic	0.959
p-value	0.641
Pass?	Yes

D. Conduct the t-test and report the results here:

t-value 112, d.f. 15, p <0.001

Do you reject the null or retain the null? Reject

E. Interpret the results – does the TLC diet reduce LDL level based on this analysis? Explain how the p-value and group means together let you reach your conclusions.

Yes, the TLC diet reduced LDL levels, because the p-value indicates a significant difference and the TLC diet has lower mean LDL levels.

F. The t-value is much larger for this analysis than for the comparison of two separate groups in question 1. Why? Hint: look at the numbers for before and after in the data sheet – is there a consistent pattern of change for each person, regardless of what their starting LDL levels were?

There is a consistent reduction in LDL for each person once they start on the TLC diet, so the standard deviation of the differences is small, which makes the t-value small, even though the degrees of freedom are cut in half.

4. Elevated cholesterol is thought to be higher in men than in women. The data to the right gives the results of a survey of 400 people, which asked their sex and whether they have high cholesterol levels.

Sex:	High	Normal	Total
Men	30	120	150
Women	20	230	250
Total	50	350	400

Answer the following probability questions based on the data in this table. You are allowed to report probabilities as fractions, but you may need convert to decimal numbers to compare probabilities.

A. What is the probability of high cholesterol in the population overall – that is $P(\text{High})$? Is this a marginal probability or a conditional probability?

$50/400 = 0.125$, which is a marginal probability

B. What is the probability, given that a person is a man, that he will have high cholesterol – that is, $P(\text{High} \mid \text{Men})$? Is this a marginal probability or a conditional probability?

$30/150 = 0.2$, which is a conditional probability

C. What is the probability that a person that is known to have high cholesterol is a man – that is, $P(\text{Men} \mid \text{High})$?

$30/50 = 0.6$

D. What is the probability of randomly selecting a person from the population that is both a man and has high cholesterol – that is $P(\text{High and Men})$?

$30/400 = 0.075$

E. Is high cholesterol independent of sex? What two probabilities did you need to compare to reach this conclusion? Note: if the probabilities you compare are not identical consider them different when you answer.

No, high cholesterol is not independent of sex. We know this because $P(\text{High})$ is 0.125 but $P(\text{High} \mid \text{Men})$ is 0.2. When the marginal and conditional probabilities of the same outcome (high LDL) are not the same the outcome is not independent of the condition variable.